Matrix Diagonalization

Theorem 1:

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

In fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are linearly independent eigenvectors of A. In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P.

In other words, A is diagonalizable if and only if there are enough eigenvectors to form a basis of $\mathbb{R}^n$. We call such a basis an **eigenvector basis** of $\mathbb{R}^n$.

Theorem 2:

An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

When A is diagonalizable but has fewer than n distinct eigenvalues, it is still possible to build P in a way that makes P automatically invertible, as the next theorem shows.

Theorem 3:

Let A be an $n \times n$ matrix whose distinct eigenvalues are $\lambda_1, \ldots, \lambda_p$.

1. For $1 \leq k \leq p$, the dimension of the eigenspace for λ_k is less than or equal to the multiplicity of the eigenvalue λ_k.
2. The matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces equals n, and this happens if and only if:
    - (i) The characteristic polynomial factors completely into linear factors.
    - (ii) The dimension of the eigenspace for each λ_k equals the multiplicity of λ_k.
3. If A is diagonalizable and $\mathcal{B}_k$ is a basis for the eigenspace corresponding to λ_k for each k, then the total collection of vectors in the sets $\mathcal{B}_1, \ldots, \mathcal{B}_p$ forms an eigenvector basis for $\mathbb{R}^n$.

Practices

Exercise 01: Determine whether the following matrices are diagonalizable. If so, find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$. For each diagonalizable matrix, write down the conclusion including the factorization $A = PDP^{-1}$ and a basis for each eigenspace. Show your work step-by-step.

a. $A_1 = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ **b.** $A_2 = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$ **c.** $A_3 = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$

d. $A_4 = \begin{bmatrix} 3 & -1 & 1 \\ 0 & 2 & 0 \\ 1 & -1 & 3 \end{bmatrix}$ **e.** $A_5 = \begin{bmatrix} 4 & 0 & -2 \\ 2 & 5 & 4 \\ 0 & 0 & 5 \end{bmatrix}$ **f.** $A_6 = \begin{bmatrix} 0 & x & 0 \\ y & 0 & 0 \\ 0 & 0 & z \end{bmatrix}$

Note: Let x, y, z be the last three digits of your student ID number (e.g., if your student ID ends in 789, then $x = 7$, $y = 8$, $z = 9$).

Exercise 02: Find all values of the parameters (x and y) for which the following matrices are diagonalizable:

a. $A_7 = \begin{bmatrix} 5 & 0 & 0 \\ 0 & -1 & x \\ 3 & 0 & y \end{bmatrix}$ **b.** $A_8 = \begin{bmatrix} 2 & 3 & x \\ 1 & 0 & -1 \\ 1 & 1 & y \end{bmatrix}$ **c.** $A_9 = \begin{bmatrix} 4 & x & -1 \\ 4 & 0 & 2 \\ 2 & -1 & y \end{bmatrix}$

Exercise 03: Based on the results obtained in **Exercise 02**, choose one specific pair of values (x, y) that makes the matrix diagonalizable for each matrix (A_7, A_8, A_9). Then, compute A^2, A^{10}, and A^{100} for those chosen diagonalizable matrices. Show your work step-by-step.